

Polygon Decompose Gallery

Unit 5: Architecture Firm • Lesson 5-4 • EduWonderLab

UNIT 5 • LESSON 5-4 • TEACHER NOTES

Standard 6.G.1

FORMAT Decompose + gallery walk	TIME 25–35 min
GROUPING Small groups	TOPIC Area of Regular Polygons
STANDARD 6.G.1	PREP LEVEL Medium — print polygon templates and feedback notes

Learning goal *I can find the area of a polygon by decomposing it into triangles and adding.*

Teacher setup

- Print one polygon set per group. Students decompose each polygon into triangles.
- They find each triangle's area, then add for the total.
- Groups post one solution for a gallery walk and give feedback.

Materials

Polygon templates, Rulers, Decomposition lines, Gallery feedback sticky notes, Pencil.

Differentiation & language support

- Pre-drawn decomposition lines help students start (SPED).
- Sentence stem: "I split the polygon into ____ triangles because ____."
- ESOL: number each triangle before finding its area.

Key vocabulary (English / Español):

Term	Español	What it means
decompose	descomponer	to break into smaller shapes
regular polygon	polígono regular	a polygon with equal sides and angles

Quick teacher check: *Check the per-triangle area used on Card 4 ($\frac{1}{2} \times 4 \times 3 = 6$).*

Polygon Decompose Gallery

Unit 5: Architecture Firm • Lesson 5-4 • EduWonderLab

Name: _____

Date: _____

Class: _____

Your goal: *I can find the area of a polygon by decomposing it into triangles and adding.*

What you need

Polygon templates, Ruler, Sticky notes.

How to play

1. Choose a polygon and draw lines to split it into triangles.
2. Find the area of each triangle.
3. Add the triangle areas for the total.
4. Post one solution and give feedback to another group.

Decomposition Cards

Break each polygon into triangles, find each area, and add to get the total.

1. A hexagon splits into 6 triangles, each 5 sq cm. Total area?	2. A pentagon splits into 5 triangles, each 6 sq cm. Total area?
3. An octagon splits into 8 triangles, each 4.5 sq cm. Total area?	4. A hexagon splits into 6 triangles, each base 4, height 3. Total area?
5. A figure splits into 4 triangles, each base 5, height 8. Total area?	6. A polygon decomposes into triangles of 9, 12, and 15 sq cm. Total?

Record your work (triangle areas + total):

Explain your thinking

Explain why decomposing a polygon into triangles makes the area easier to find.

Sentence frame: *I split the polygon into ___ triangles because ___.*

★ **Challenge:** A regular hexagon decomposes into 6 equal triangles. If the total area is 48 sq cm, what is each triangle's area? (8 sq cm)

ANSWER KEY

Teacher copy — please do not distribute to students.

UNIT 5 • LESSON 5-4 • Polygon Decompose Gallery

Standard 6.G.1

Answers

1. A hexagon splits into 6 triangles, each 5 sq cm. Total area? Answer: 30 sq cm	2. A pentagon splits into 5 triangles, each 6 sq cm. Total area? Answer: 30 sq cm
3. An octagon splits into 8 triangles, each 4.5 sq cm. Total area? Answer: 36 sq cm	4. A hexagon splits into 6 triangles, each base 4, height 3. Total area? Answer: 36 sq cm (6×6)
5. A figure splits into 4 triangles, each base 5, height 8. Total area? Answer: 80 sq cm (4×20)	6. A polygon decomposes into triangles of 9, 12, and 15 sq cm. Total? Answer: 36 sq cm

Teaching notes & look-fors

- Total area = sum of all triangle areas.
- Challenge: $48 \div 6 = 8$ sq cm per triangle.